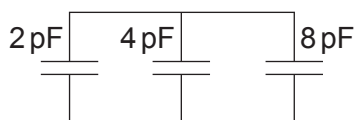


SECTION AAnswer **all** questions.

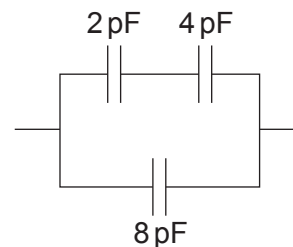
1. (a) Calculate the total capacitance of each of the following capacitor combinations. [6]



(i)



(ii)



(iii)

(i)

.....

.....

(ii)

.....

.....

(iii)

.....

.....



- (b) Michael, a laboratory technician, must make a $2.7\ \mu\text{F}$ capacitor using only two square sheets of aluminium of side 42.0 cm. Determine whether or not it is realistic for Michael to produce a capacitance of $2.7\ \mu\text{F}$ using these two sheets of aluminium and no dielectric. [4]

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- (c) State what happens to the capacitance of a parallel plate capacitor when a dielectric is placed between the plates. [1]

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